

Rajat Tiwari

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PROFESSIONAL SUMMARY

Data Scientist with 3.5+ years of experience building machine learning, NLP, and computer vision solutions using Python and SQL. Experienced in data preprocessing, feature engineering, predictive modeling, and transformer-based NLP models. Hands-on experience with autonomous driving datasets, embedded sensor analytics, and end-to-end ML pipelines. Strong foundation in AWS, Agile development, and translating business problems into data-driven solutions.

EDUCATION

Pace University, Seidenberg School of Computer Science and Information Systems New York, NY
Master of Science (MS) in Data Science | **GPA:** 3.89/4 Dec 2025

Pune University, Dr. D. Y. Patil, School of Engineering Pune, India
Bachelor of Engineering (B.E) in Computer Engineering | **CGPA:** 8.63/10 July 2023

PROFESSIONAL EXPERIENCE

Community Dreams Foundation—New Jersey, USA

Data Scientist Feb 2024 – Present

- Built predictive models and analytics solutions using Python and SQL to support community programs.
- Developed NLP pipelines for feedback and survey text classification.
- Cleaned, transformed, and analyzed structured and unstructured datasets.
- Created Power BI dashboards to monitor KPIs and program performance.
- Automated reporting workflows, reducing manual effort by **40%**.

KPIT Technologies — Bangalore, India

Data Scientist Nov 2022 – Dec 2023

- Analyzed real-time sensor data collected from **Arduino Uno** and **Raspberry Pi** systems using Python for embedded AI applications.
- Built data preprocessing pipelines to clean, transform, and validate sensor datasets for machine learning workflows.
- Performed exploratory data analysis (EDA) and feature engineering to improve predictive modeling on automotive datasets.
- Collaborated with embedded software engineers to integrate data-driven insights into intelligent vehicle systems.
- Developed dashboards and reports to monitor model performance and operational metrics.

Ideas to Impacts — Pune, India

Data Scientist Nov 2020 – Nov 2022

- Prepared high-quality training datasets for autonomous driving AI by annotating video frames and assigning hierarchical priorities to road objects.
- Performed quality assurance and validation of labeled datasets to improve computer vision model accuracy.
- Built preprocessing workflows to standardize image metadata and annotation outputs for ML pipelines.
- Conducted exploratory analysis on annotated datasets to identify labeling inconsistencies and improve dataset quality.

- Collaborated with ML engineers to support computer vision model training, evaluation, and continuous dataset improvement.

PROJECTS

Capstone Project — Early Detection of Foodborne Illness Outbreaks (NYC)

Sept 2025 – Dec 2025

Python | NLP | DistilBERT | Scikit-learn | Time Series

- Built an end-to-end ML pipeline using Reddit, Yelp, restaurant inspection, and outbreak datasets.
- Developed a TF-IDF + Logistic Regression baseline and fine-tuned DistilBERT for illness text classification.
- Applied time-series analysis to identify outbreak trends and anomalies.
- Evaluated models using Precision, Recall, F1-score, and ROC-AUC.
- Created interactive dashboards to visualize hotspot detection and outbreak insights.

Autonomous Driving Data Intelligence System

Ideas to Impacts | Python | Computer Vision | Data Annotation | OpenCV

- Prepared high-quality training datasets by annotating video frames for autonomous driving AI models.
- Assigned hierarchical priorities to road objects including vehicles, pedestrians, traffic signs, and lane markings.
- Performed quality assurance on annotated data to improve computer vision model accuracy.
- Built preprocessing workflows to standardize annotation outputs for ML training.
- Collaborated with ML engineers to support object detection model development and validation.

Embedded Sensor Analytics for Intelligent Vehicle Systems

KPIT Technologies | Python | Raspberry Pi | Arduino Uno | Machine Learning

- Collected and analyzed real-time sensor data from embedded systems for automotive applications.
- Built preprocessing and feature engineering pipelines for structured sensor datasets.
- Performed exploratory data analysis to identify patterns and optimize system performance.
- Developed predictive models using Python to support intelligent embedded applications.
- Collaborated with embedded software engineers to integrate data-driven insights into automotive solutions.

SKILLS

Programming: Python, SQL, C++

Machine Learning: Scikit-learn, Regression, Classification, Feature Engineering, Predictive Modeling

Generative AI & NLP: BERT, DistilBERT, Prompt Engineering, Text Classification, TF-IDF, LLM Evaluation

Computer Vision: Data Annotation, Dataset Preparation, Image Data Preprocessing, Object Detection

Data Analysis: Exploratory Data Analysis (EDA), Data Cleaning, Data Preprocessing, Statistical Analysis

Data Visualization: Power BI, Matplotlib, Pandas, NumPy, Excel

Cloud & Tools: AWS (S3, EC2, IAM), Git, Jupyter Notebook, Docker (Basic), CI/CD Concepts

Embedded & IoT: Raspberry Pi, Arduino Uno, Sensor Data Analysis

Methodologies: Agile, SDLC, Model Experimentation, ML Pipeline Development

PUBLICATIONS

Patient Information Retrieval Using Fingerprint,” *International Journal of Scientific Research in Science, Engineering and Technology (IJSRSET)*, Print ISSN : 2395-1990, [Online ISSN : 2394-4099](#), Volume 10, Issue 8, pp. 26-32, May-June-2023.